



CYBERSECURITY AND AI CASE STUDIES
(MOD006570 – Element 010 Coursework Report)

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GitHub: <https://github.com/vinaykoparthi-cpu/Cyber-Security.git>

UNSW-NB15 Based Intrusion Detection and Insider Label Poisoning

1. AI model selection and intrusion detection outcomes.

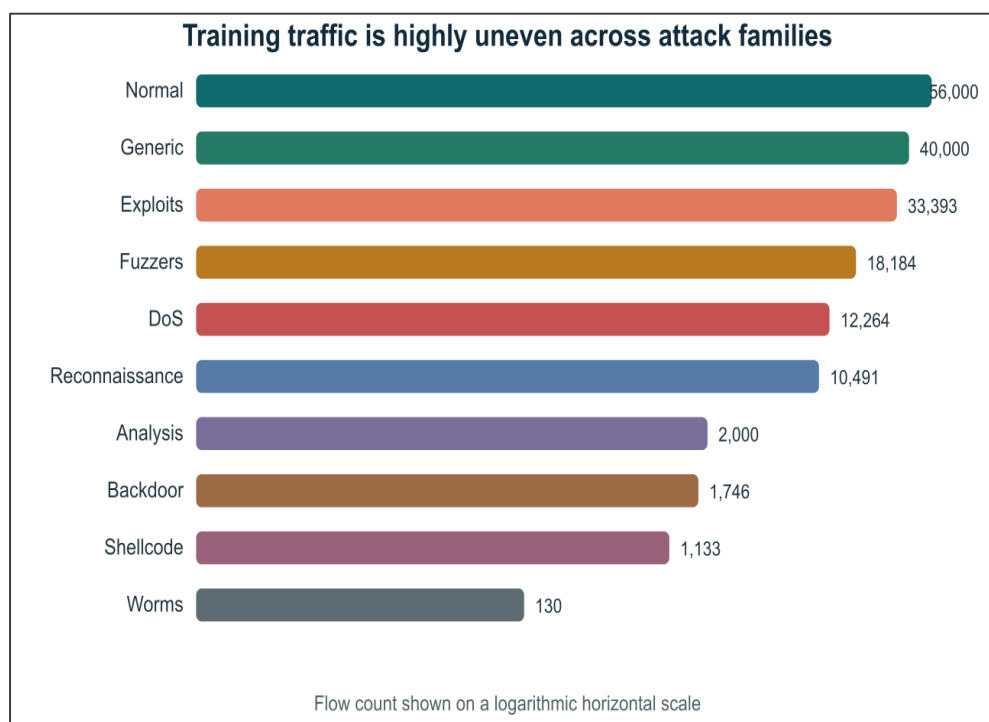
1.1 Problem description and dataset

The task was binary intrusion detection - label 0 was regular traffic and label 1 was malicious traffic. The official UNSW-NB15 split provided 175,341 training and 82,332 test flows over nine attack families. The test set was kept for final examination. The UNSW-NB15 dataset contains recent normal and attack behaviour and is commonly utilised for network-intrusion research (Moustafa and Slay, 2015; Moustafa and Slay, 2016). Checks revealed no duplicate, missing cell or schema mismatch, supporting consistent train-test comparison.

1.2 Exploratory analysis

Normal, Generic and Exploits are the majority of training data, Worms includes only 130 samples. Therefore, the log-scaled distribution shows imbalance that can be hidden by accuracy alone. Overlapping byte patterns were observed in robust summaries and density graphs within the notebook, which validated a multifeatured nonlinear detector and necessitated examining recall by attack family.

Figure 1. Distribution of UNSW-NB15 training flows by traffic category (logarithmic scale).



1.3 Preprocessing and leak control

Removed fields: identifier, attack cat. The attack cat would directly give away the solution. Numeric variables were imputed using median and standardised, while proto, service and state were imputed using mode and one-hot encoded. To avoid leaking (Pedregosa et al., 2011), a pipeline was fit on the training split only, each transformation was applied. A stratified 80:20 train-validation split was employed for model selection.

1.4 Model Selection Experiment

A class weighted logistic regression yielded a linear baseline. We tried three MLPs with one versus two hidden layers and enhanced L2 regularisation. All models employed the same pre-processing; MLPs used ReLU, early stopping and seed 42. Attack F1 was the metric of choice, with recall, balanced accuracy, AUC, false alarms and fitting time supplied for context.

Table 1. Validation results used to select the intrusion-detection model.

Model	Attack precision	Attack recall	Attack F1	Balanced accuracy	ROC AUC	Fit
Wide MLP (64, 32)	0.9502	0.9739	0.9619	0.9325	0.9904	37.47
Small MLP (32)	0.9479	0.9742	0.9608	0.9300	0.9893	9.84
Regularised MLP (64, 32)	0.9485	0.9700	0.9591	0.9288	0.9884	8.36
Linear baseline	0.9502	0.9509	0.9506	0.9223	0.9845	4.19

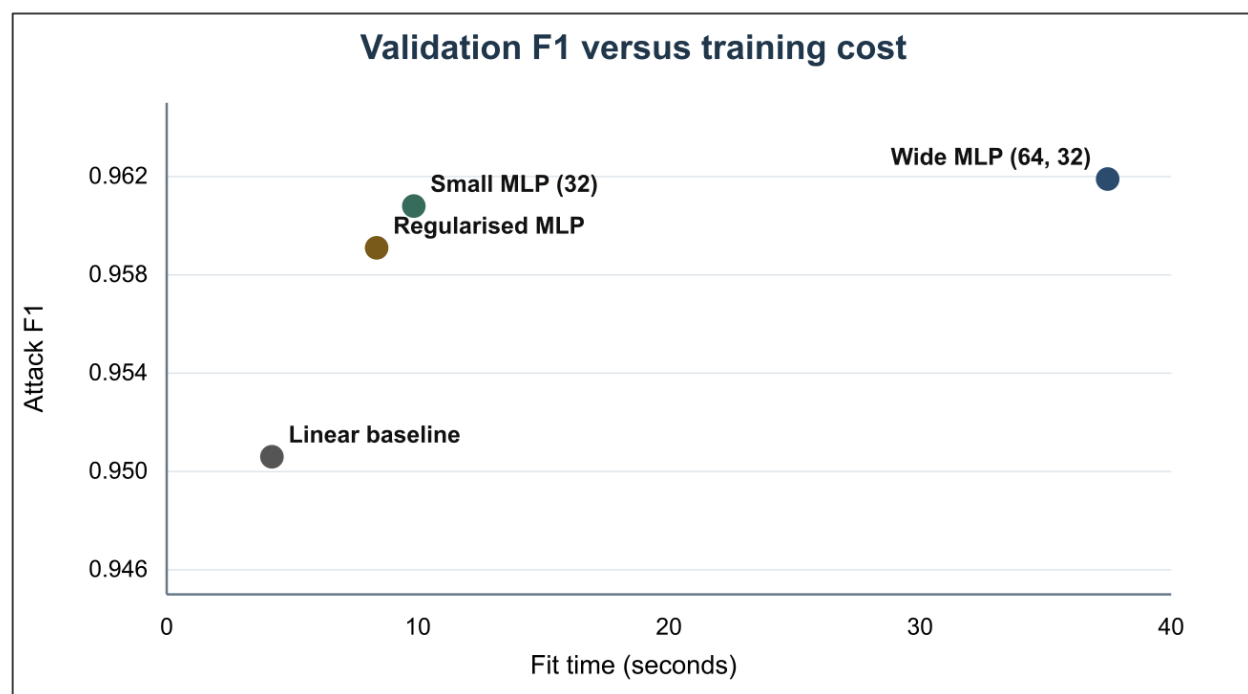


Figure 2. Validation attack F1 plotted against model fitting time.

The Wide MLP achieved the highest validation F1 (0.9619), balanced accuracy (0.9325) and AUC (0.9904), experimentally justifying the nonlinear modelling. The Small MLP was just 0.0011 lower

in F1 and substantially faster, thus would be good for a restricted gateway. The Wide MLP was preserved with the maximal validation attack F1 as the declaration selection criteria.

1.5 Final clean-model Results.

The Wide MLP retrained on all training flows has test precision of 0.8229, recall of 0.9781, F1 of 0.8938, balanced accuracy of 0.8601 and AUC of 0.9783. The detector missed 2.19% of attacks and misclassified 25.78% of normal flows. False warnings can fatigue analysts or quarantine innocent devices on an IoT network. Missed attacks keep hacked devices in operation. The discrepancy between validation and test F1 also shows distribution change. Hence, deployment would involve local threshold calibration, an explicit false alarm budget and constant monitoring, rather than simply accepting the default 0.5 threshold

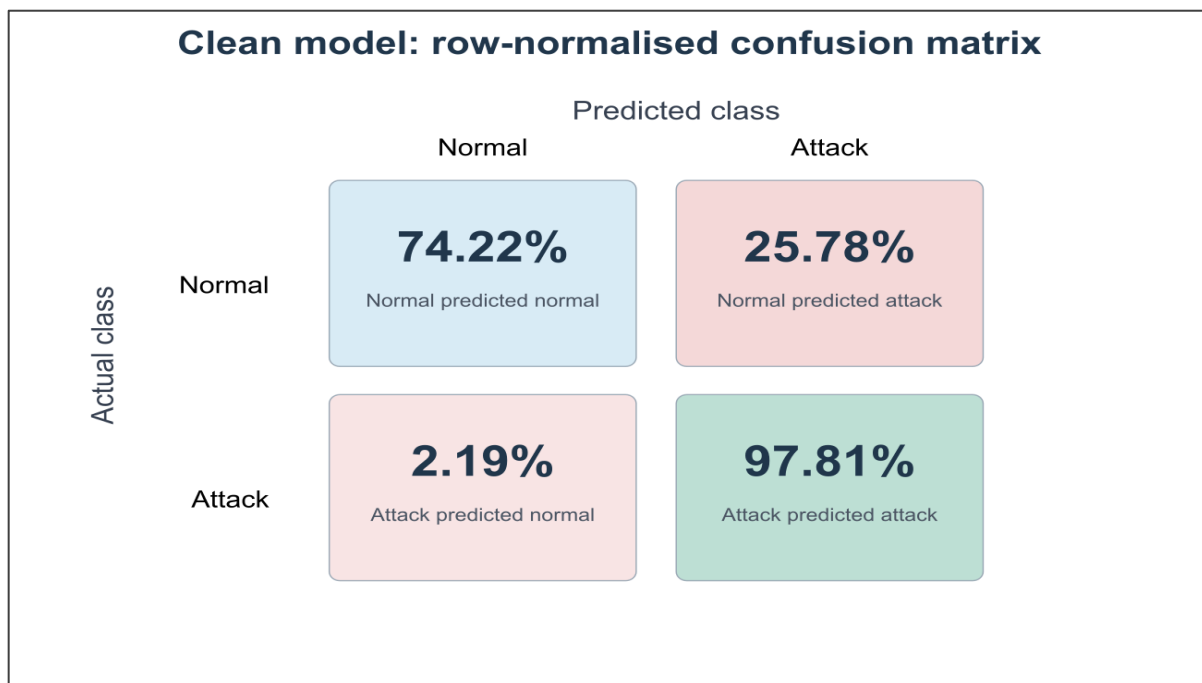


Figure 3. Row-normalised confusion matrix for the clean Wide MLP.

2. Insider Label Poisoning Attack

2.1 Threat modelling and evaluation plan.

The insider is allowed to change training labels but not features or the untouched test set. This is an integrity attack in the training phase that makes syntactically valid labels incongruous with their flows. Carefully picked poisoning sites can shift a decision boundary (Biggio, Nelson and Laskov, 2012) and this is classified by NIST as adversarial data poisoning (Vassilev et al., 2025). The same Wide MLP pipeline was retrained at 0%, 5%, 10%, 15%, 20% and 25% poisoning independently and tested against identical genuine test labels.

2.2 Poisoning Attacks.

Random flipping reverses a seeded sample of labels, modifying both attack and normal flows. It is a good noise baseline but wastes part of the attacker's budget on adjustments that do not mask threats. The targeted method ordered true training attacks by the attack probability of the clean MLP, then flipped the attacks with the least confidence from 1 to 0. These boundary situations provide more leverage, and every label modification is orientated towards the goal of masking attacks. The method is consequently more budget-aware than random flipping and follows studies into selective poisoning from label sanitisation (Paudice, Muñoz-González and Lupu, 2019).

2.3 Results

Table 2. Performance on the untouched test set after training-label poisoning.

Poisoning	Changed	Attack precision	Attack recall	Attack F1	False-alarm rate	Miss rate
Clean 0%	0	0.8229	0.9781	0.8938	0.2578	0.0219
Random 5%	8,767	0.8031	0.9809	0.8832	0.2946	0.0191
Random 10%	17,534	0.8098	0.9772	0.8857	0.2812	0.0228
Random 15%	26,301	0.7811	0.9829	0.8704	0.3375	0.0171
Random 20%	35,068	0.7903	0.9867	0.8776	0.3208	0.0133
Random 25%	43,835	0.7877	0.9732	0.8707	0.3213	0.0268
Targeted 5%	8,767	0.8936	0.9355	0.9141	0.1364	0.0645
Targeted 10%	17,534	0.9621	0.8685	0.9129	0.0419	0.1315
Targeted 15%	26,301	0.9945	0.7878	0.8792	0.0053	0.2122
Targeted 20%	35,068	0.9953	0.7053	0.8256	0.0041	0.2947
Targeted 25%	43,835	0.9958	0.6484	0.7854	0.0033	0.3516

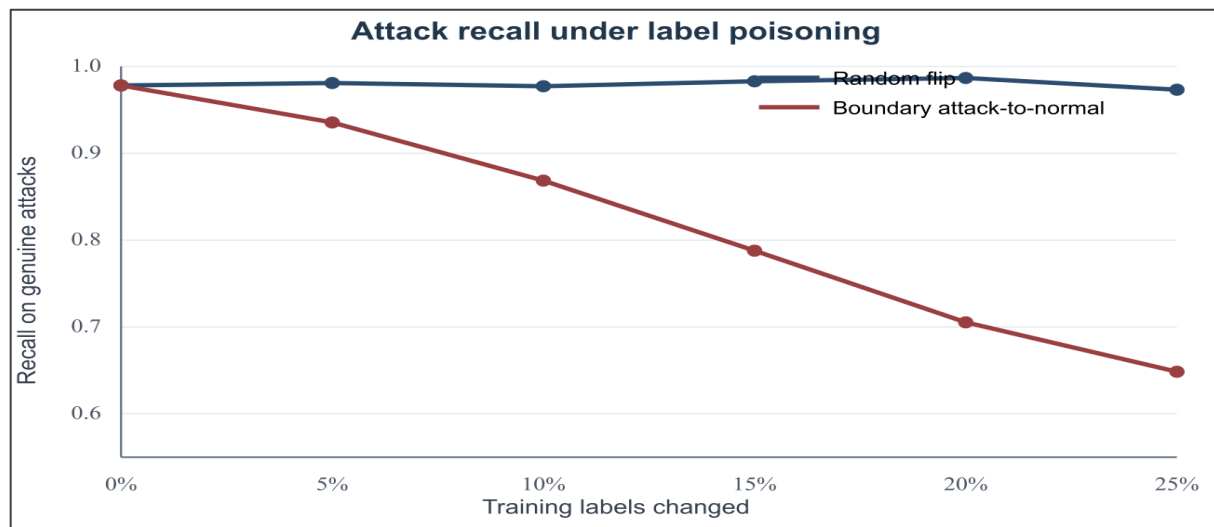


Figure 4. Recall of attacks for random and boundary-targeted label poisoning.

2.4 Critical investigation of the consequences of poisoning

Random flipping decreased recall from 0.9732 to 0.9867 but increased operational noise. The precision dropped to 0.7811, and false-alarm rate increased to 0.3375 at 15%. At 5% recall dropped

from 0.9781 to 0.9355 and the miss rate nearly tripled. Recall at 10% was 0.8685. At 25%, recall was 0.6484 and the model missed 35.16% of real attacks: around 15,939 attacks compared to 993 for the clean model. At 25% precision increased to .9958 and false alarms decreased to .0033, but this was not improvement.

At 5% all approaches altered 8,767 labels but focused poisoning caused around 1,931 more misses whereas random flipping modestly improved recall. The emphasis on uncertain assaults optimised damage per edit. The result shows the low-budget failure technique needed by the assignment, but with one seed and a 25% maximum limit broad statistical or out-of-range claims.

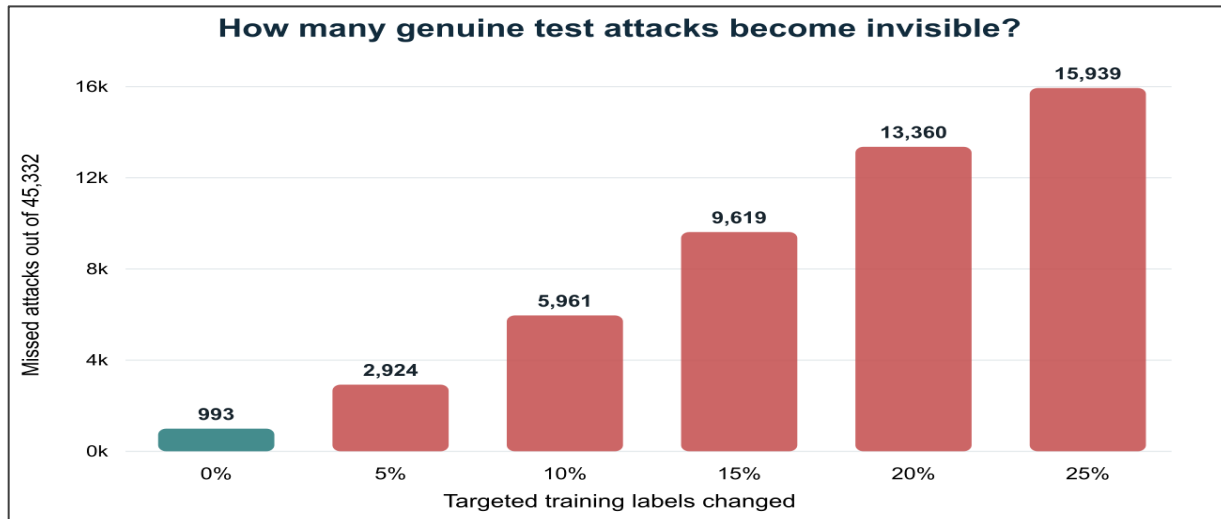


Figure 5. Number of genuine attacks missed at each targeted poisoning level.

2.5 Unequal impact across attack families

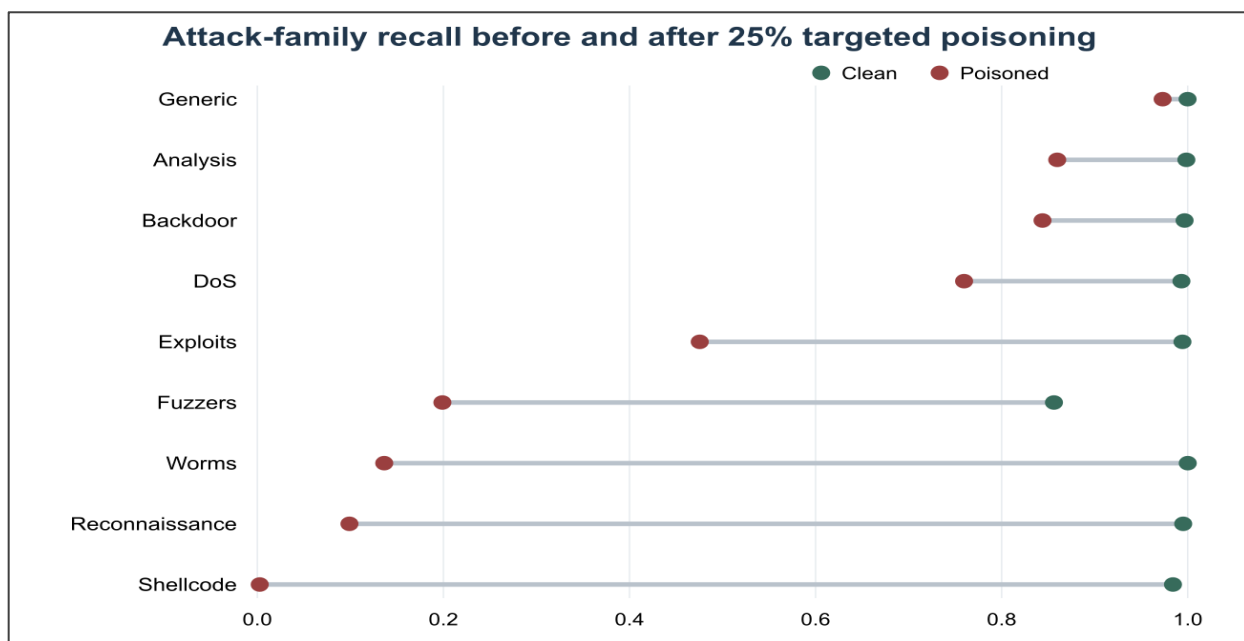


Figure 6. Clean and poisoned recall by attack family after 25% targeted poisoning.

Family-level injury was heterogeneous. At 25% targeted poisoning, generic recall was still 0.9730 but Shellcode was reduced to 0.0026, Reconnaissance to 0.0990, Worms to 0.1364 and Fuzzers to 0.1989. Thus, a global metric can mask the loss of entire threat behaviours

2.6 Implications and constraints for defence

Defences should include immutable versions of datasets, authenticated label modifications, use of least privilege, monitoring of transition history and a protected validation set. Before signing a model, promotion gates should examine miss rate and min family remember. Signed artefacts and rollback points restrict damage after deployment, whereas boundary samples demand review or sanitisation. The attack presupposes a Whitebox access to the clean pipeline. Repeated seeds and confidence intervals will improve the random baseline.

3. Architecture for Secure Deployment of Autonomous Agent.

3.1 Security principles in literature

Its architecture decouples responsibilities for data, model-release and live-operation. This is in line with multi-agent advise on defined responsibility boundaries (Jennings, 2000) and zero-trust guidance which requires explicit service authentication instead of trusting network location (Rose et al., 2020).

3.2 Block diagram of the whole system

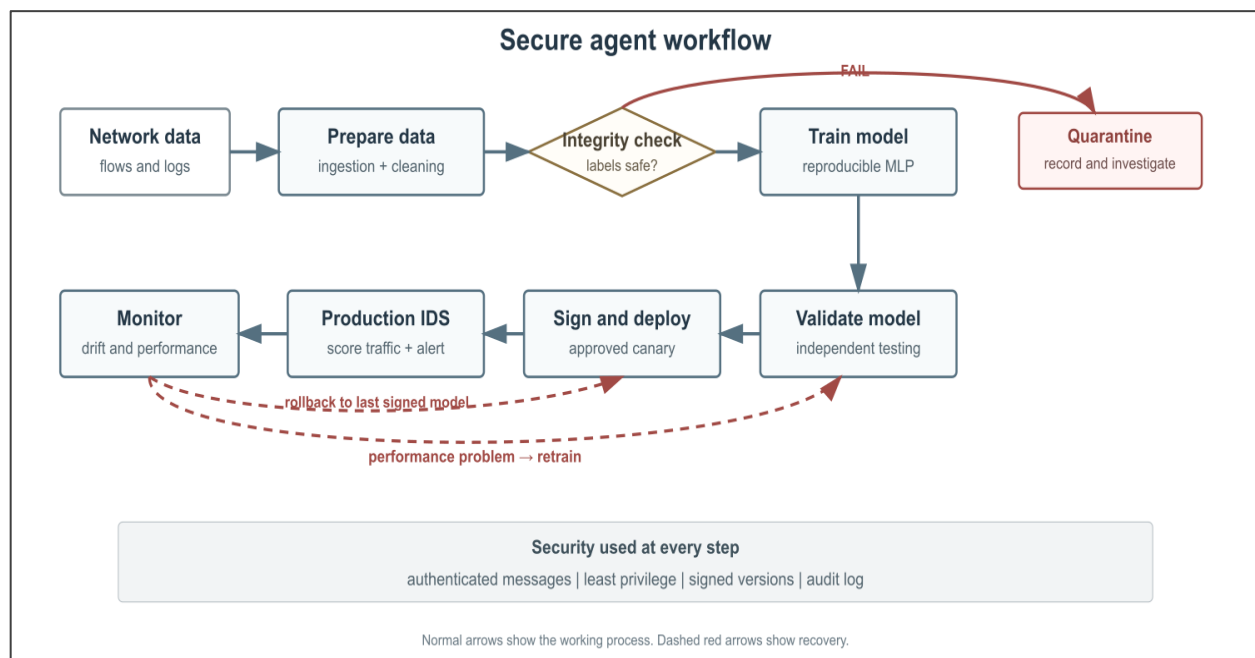


Figure 7. Proposed secure autonomous-agent architecture for trustworthy model deployment.

3.3 Explanation of diagram and agent duties

1. Cleaning agents and ingestion collect trusted network data and provide versioned features.
2. The integrity agent verifies the labels and puts suspect data into quarantine before the training
3. The training agent constructs the MLP but does not have the power to approve or deploy it.
4. The validation agent checks recall, false alarms and poisoning resistance independently.
5. Only a validated model gets signed and released via a controlled canary deployment.
6. The production IDS scores the traffic, while the health monitor looks for drift and performance.

References

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